Adam Tang

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Available Spring + Summer 2027 (up to 8 months, co-op or internship)

Engineering Portfolio

antangled.github.io/engineering_portfolio – code repos, CAD, build photos, and test videos for every project below

Education

University of California, Berkeley Expected May 2028
B.S. Electrical Engineering and Computer Sciences, Minor in Bioengineering | **GPA: 3.91/4.00** *Berkeley, CA*
- **Honors:** Tau Beta Pi and Eta Kappa Nu (IEEE-HKN) engineering honor societies, Fall 2026
- **Coursework:** EECS 16B (circuits, LTspice), EECS 16A (signals), CS 61C (computer architecture, C), CS 61A (Python)

Projects

Kinema: 6-DOF Camera Arm, Custom Cycloidal Actuators | *Self-initiated, SolidWorks, FEA* June 2026 – Present
- Designed a 10:1 internal cycloidal actuator packaged coaxially inside a frameless BLDC (10.4 N·m peak) by running tolerance stack-ups on the cycloid and pin fits, selecting bearings, and validating the housing with FEA
- Built 7+ CNC-machined and 3D-printed prototypes, then brought up the motor driver and absolute encoder for closed-loop position control; bench-tested to 2 N·m, earning a CubeMars sponsorship and $1.5k Jacobs Ignite grant

PolySynth X: MPE Keyboard and Synth | *Self-initiated, STM32H7, C++, KiCad* Nov. 2025 – June 2026
- Built a working 12-key MPE keyboard with an onboard synth engine by iterating compliant hammer-action keys 7+ times in Onshape and 3D print for touch accuracy, and designing hall-effect sensor PCBs in KiCad over 3+ revisions
- Wrote real-time C++ DSP firmware on STM32H7 (Daisy Seed) that differences two hall-effect sensors under each key via ADC, giving every key independent pitch bend

Modular Biodiversity Sensing Module (Hardware Lead) | *ESP32-S3, Solar Power, Edge AI* Oct. 2025 – June 2026
- Designed a weatherproof, fully solar-powered sensor node (buck-converter power system, camera, acoustic, and thermal sensors) running an on-device AI model that identifies species with confidence scores; first node deployed in the field

Compact Automated Aquaponics System | *Self-initiated, Fusion 360, Sensors* Dec. 2023 – June 2025
- Hand-built an aquaponics system with a sensor and control stack monitoring fish and plant health; deployed on campus

Envoy: AI Client-Outreach Automation | *Hackathon Winner* Oct. 2026
- Automated client outreach with AI; won the AI Enterprise Track at Berkeley SCET's Innovation Intelligence Hackathon

Experience

Embedded Systems and Hardware Integration Intern Dec. 2025 – Aug. 2026
Eldaeon (RF sensing startup) *Berkeley, CA*
- Built a real-time drone and UAP detection pipeline in Python on Raspberry Pi that cross-correlates RF Doppler returns with aircraft ADS-B and ship AIS tracks, flagging uncorrelated targets on a live dashboard with phone alerts via AWS and Twilio
- Designed the antenna pan-tilt mechanism and wrote calibration firmware that sweeps $\pm 10^{\circ}$ to lock onto the strongest local transmitter, yielding the clearest Doppler signal; field-tested across 3+ outdoor campaigns and an SF boat-detection run
- Mapped each component's power budget from datasheets and wrote a power-sequencing daemon (DPOL) that keeps servo moves and SDR bursts from overlapping, cutting projected draw 30.2% to under the 15.4 W PoE limit

Solar Engineering Project Manager Sep. 2025 – May 2026
CalSol, UC Berkeley Solar Racing Team *Berkeley, CA*
- Designed the solar array and bypass-diode layout so the battery keeps charging even when panels are partially shaded, a projected +12% partial-shade efficiency gain
- Designed MPPT charge controller enclosures in SolidWorks and integrated them into the wiring layout with the batteries

Technical Project Manager and Systems Modeling Engineer Feb. 2026 – Present
BEACN (Bay Area Environmentally Aware Consulting Network) *Berkeley, CA*
- Lead a 5-person team evaluating LFP, NMC, and sodium-ion battery storage for GridPoint's multi-hundred-site rollout and building an AI/ML roadmap to speed site scoring, interconnection, and permitting
- Built statistical energy model and GIS analysis for AI data center firm Vigilent, setting efficiency targets vs. PUE

Mechanical Lead Nov. 2021 – June 2024
MUREX Exeter, MATE Marine Robotics Competition Team *Exeter, NH*
- Led a 6-person team building a 6-DOF underwater ROV and 7-DOF manipulator; cut manufacturing cost 88% by moving outsourced parts to in-house CNC machining and 3D printing; ran design reviews; 6th globally at MATE 2024, 10 records

Leadership

Founder and President Sep. 2020 – June 2025
Innovation Towards the Future (climate NGO) *Potomac, MD*
- Founded and led a climate NGO for 5 years, earning admission as a UNFCCC observer organization and securing 3+ corporate partnerships; selected as USA youth delegate to UN COP28 in Dubai from 5k applicant NGOs

Technical Skills

Mechanical: SolidWorks, Fusion 360, Onshape, FEA, tolerance stack-ups, actuator and compliant mechanism design, DFM/DFA
Electrical: KiCad PCB design, board bring-up and validation, analog signal conditioning, BLDC drivers, encoders, SDR, MPPT
Embedded and Software: C++, Python, Java, Verilog, STM32H7, ESP32-S3, Raspberry Pi, ADC, real-time DSP, AWS, Twilio
Fabrication and Test: Rapid prototyping, CNC machining, 3D printing, assembly, bench and field testing, power budgeting